

Shihan Kanungo

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Palo Alto, California • Birthdate: 01/08/2010 • US Citizen

EDUCATION

Massachusetts Institute of Technology (MIT) Mathematics major; minor in Music	2026 — 2030 (expected)
San José State University (SJSU) Concurrent enrollment during high school. GPA: 4.0. Coursework listed on page 3	2023 — 2026
Euler Circle College-level math courses taught by Simon Rubinstein-Salzedo. Coursework listed on page 4	2019 — 2026

SELECTED HONORS AND AWARDS

The William Lowell Putnam Mathematical Competition Ranked 22nd (N2) in 2025 — as a 15 yo. in high school. Ranked 58 in 2024.	2024 — 2025
17th Romanian Master in Mathematics, RMM 2026. Member of USA team for RMM 2026, Feb 23–28, Bucharest. Bronze medal.	2026
USA Mathematical Olympiad (USAMO) Gold medal.	2025, 2026
MAA's Mathematical Olympiad Program (MOP) Attended 'Blue' MOP. Selected for USA TST group for IMO 2026.	2025
USA Physics Olympiad (USAPhO) Gold medalist and member of 2024 USA Physics team.	2024
USA Mathematical Talent Search (USAMTS) 2× Gold medal and top scorer both times — 2025 (75/75) and 2023 (74/75).	2023, 2025
Shing-Tung Yau High School Science Award Global research competition. Won (with S. Erat) the Bronze medal in Mathematics (North America).	2024
Spirit of Ramanujan Fellow, World Science Scholar, 2025 Global Child Prodigy Selected fellowships and awards.	2023 — 2025

RESEARCH PUBLICATIONS AND PREPRINTS

Upper bounds for sequence saturation. PRIMES-USA 2025: Research project in combinatorics; Adviser: Dr. Jesse Geneson (SJSU) S. Kanungo. <i>Electronic Journal of Combinatorics</i> (In review). [arXiv]	2025
Representation theory of the twisted Yangians in complex rank. PRIMES-USA 2024: Research project in representation theory; Adviser: Dr. Arun S Kannan A. Kannan, S. Kanungo. <i>Algebras and Representation Theory</i> (In review). [arXiv]	2024
Mixed Tensor Products, Capelli Berezinians, and Newton's Formula for $\mathrm{gl}(m n)$. PRIMES-USA 2024: Research project in representation theory; Adviser: Dr. Arun S Kannan S. Erat, A. Kannan, S. Kanungo. <i>Transformation Groups</i> (2025). [arXiv] [journal] [PDF]	2024
On Product Formulas of Guillera and Sondow. Research project in number theory; Adviser: Dr. Jordan Schettler (SJSU) S. Kanungo, J. Schettler. <i>The Ramanujan Journal</i> (In review). [arXiv]	2024
A weaker notion of the finite factorization property. PRIMES-USA 2023: Research project in commutative algebra; Adviser: Dr. Felix Gotti H. Jiang, S. Kanungo , H. Kim. <i>Commun. Korean Math. Soc.</i> 39 (2024), No. 2, pp. 313-329. [arXiv] [journal] [PDF]	2023

EXPOSITORY PAPERS

Part of the requirements for courses at Euler Circle and graduate courses at SJSU

[arXiv]	Elliptic Functions and Eisenstein Series.	Complex Analysis, Number Theory	2026
[arXiv]	The Grothendieck Group and K-Theory.	Algebra, Algebraic Topology	2026
[PDF]	Height, Krull Dimension, and Localization.	Commutative Algebra	2025
[PDF]	Modular Curves as Moduli Spaces.	Algebraic Geometry	2025
[PDF]	The Fundamental Group.	Algebraic Topology	2025
[PDF]	On the S-matrix Conjecture for even n.	Nonnegative Matrices	2025
[PDF]	David Gale's Subset Takeaway Game.	Combinatorial Game Theory	2024
[PDF]	Chromatic Polynomials.	Graph Theory	2024
[PDF]	Brun's Theorem.	Number Theory	2024
[PDF]	The Stochastic Gradient Descent Method.	Stochastic Methods	2025
[arXiv]	Sobolev Spaces and Applications to PDEs.	Theory of PDEs	2025
[arXiv]	Space-Filling Curves.	Real Analysis	2024
[PDF]	The Hahn-Banach Theorem and Applications.	Functional Analysis	2025
[PDF]	Lagrangian and Hamiltonian Mechanics.	Dynamical Systems	2024
[PDF]	Dynamical Billiards.	Ergodic Theory	2024
[PDF]	Large Gaps Between Primes.	Analytic Number Theory	2024
[PDF]	Arithmetic Dynamics.	Number Theory	2023
[PDF]	Minimal surfaces.	Differential Geometry	2022
[PDF]	Herbert Wilf's Snake-Oil Method.	Generating Functions	2021
[PDF]	Mixing-time estimates for riffle shuffle.	Markov Chains	2020
[PDF]	Classifying groups of certain orders.	Abstract Algebra	2020

SELECTED CONFERENCES AND SUMMER PROGRAMS

MAA's Mathematical Olympiad Program (MOP).	2025
AAPT's US Physics Team Training Camp.	2024
AMS Joint Mathematics Meetings (JMM).	2023, 2024
MIT PRIMES Fall Conference.	2023, 2024, 2025
The Special Lectures at Kharkiv-Vienna International Science School.	2023
Summer Workshop for Intrepid Mathematicians.	2023

TECHNOLOGIES

Python, Sage, Wolfram Alpha, $\text{\LaTeX}$, Asymptote, Java, C++, Manim, HTML

Coursework

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San José State University

Concurrent enrollment during high school. Graduate courses indicated by †

A+	MATH 279A † Graph Theory. Yan Zhang	Spring 2024
	Douglas B. West. <i>Introduction to Graph Theory</i> . Prentice Hall 1996	
A	MATH 221A † Higher Algebra I. Edgar Bering	Fall 2025
	Paulo Aluffi. <i>Algebra: Chapter 0</i> . AMS 2021. Covered Chs. 1–4.	
A+	MATH 221B † Higher Algebra II. Edgar Bering	Spring 2026
	Paulo Aluffi. <i>Algebra: Chapter 0</i> . AMS 2021. Covered Chs. 5–9.	
A+	MATH 226 † Theory of Numbers. Jordan Schettler	Spring 2024
	Primes in arithmetic progressions, partitions, modular group, and the Dedekind eta function.	
A+	MATH 285M † Elliptic Curves and Modular Forms. Jordan Schettler	Fall 2025
	Alvaro Lozano-Robledo. <i>Elliptic Curves, Modular Forms, and Their L-functions</i> . AMS 2011.	
A+	MATH 229 † Advanced Matrix Theory. Wasin So	Fall 2024
	R. Horn & C. Johnson. <i>Matrix Analysis</i> , Cambridge University Press 2012.	
A+	MATH 285M † Nonnegative matrices. Wasin So	Spring 2025
	The basic theory of nonnegative matrices; applications to graphs and digraphs.	
A+	MATH 275A † Topology. Yan Zhang	Fall 2025
	Compactness, connectedness; Tychonoff's Theorem; paracompactness and metrization theorems.	
A+	MATH 231A † Real Analysis. Slobodan Simic	Fall 2024
	E.M.Stein & R.Shakarchi, <i>Real Analysis: Measure Theory, Integration, and Hilbert Spaces</i> , Chs. 1–4.	
A+	MATH 231B † Functional Analysis. Slobodan Simic	Spring 2025
	L Debnath & P Mikusinski. <i>Introduction to Hilbert Spaces with Applications</i> . Academic Press 2005.	
A+	MATH 238 † Advanced Complex Variables. Jordan Schettler	Spring 2026
	E.M.Stein & R.Shakarchi, <i>Complex Analysis</i> , Chs. 1–5, 8.	
A+	MATH 233A † Partial Differential Equations. Daniel Brinkman	Spring 2025
	Peter J Olver. <i>Introduction to Partial Differential Equations</i> . Springer 2013.	
A+	MATH 234 † Advanced Dynamical Systems. Liam Stanton	Spring 2024
	Continuous and discrete systems; stability of equilibria and closed orbits, structural stability.	
A	MATH 285A † Stochastic Methods in Scientific Computing. Liam Stanton	Spring 2025
	Monte Carlo methods, Metropolis-Hastings algorithm, random-number generation.	
A	MATH 129A Linear Algebra I. Dashiell Fryer	Fall 2023
	S.H.Friedberg, et al. <i>Linear Algebra</i> 2nd ed., Prentice Hall 1989.	
A+	MATH 129B Linear Algebra II. Wasin So	Spring 2024
	S.H.Friedberg, et al. <i>Linear Algebra</i> 2nd ed., Prentice Hall 1989.	
A+	MATH 131A Introduction to Analysis I. Tim Hsu	Spring 2024
	Kenneth Ross, <i>Elementary Analysis: The Theory of Calculus</i> . UTM, Springer NY 2013.	
A+	MATH 131B Introduction to Analysis II. Tim Hsu	Fall 2024
	Tim Hsu, <i>Fourier Series, Fourier Transforms, and Function Spaces</i> . AMS/MAA 2020	
A+	MATH 138 Complex Variables. Liam Stanton	Fall 2023
	Dennis G. Zill et al, <i>A First Course in Complex Analysis with Applications</i> . Jones & Bartlett 2006.	
A+	MATH 179 Introduction to Graph Theory. Sogol Jahanbekan	Fall 2024
	R. J. Wilson. <i>Introduction to Graph Theory</i> . Prentice Hall 2010	

Euler Circle

College-level math classes taught by Simon Rubinstein-Salzedo. Grades not assigned.

Abstract Algebra.

Spring 2020

Group theory, Sylow theorems, fields and extensions, number fields, Galois correspondence.

Ring Theory and Algebraic Geometry .

Summer 2026

Affine and projective algebraic geometry, Hilbert's Nullstellensatz, localization and local rings, affine schemes and morphisms.

Combinatorics.

Fall 2025

Special numbers; sieving; restricted permutations; posets; incidence algebras; generating functions.

Combinatorial Game Theory.

Fall 2024

Nim, Hackenbush, surreal numbers, impartial games, Sprague-Grundy theory.

Generating Functions.

Fall 2021

Philippe Flajolet and Robert Sedgewick. *Analytic Combinatorics*. Cambridge University Press 2011.

Number Theory.

Fall 2023

Reciprocity theorems, quadratic forms, elliptic curves, modular curves.

Analytic Number Theory.

Spring 2024

Dirichlet's Theorem, The Prime Number Theorem, Brun's theorem, smooth numbers.

Complex Analysis I.

Spring 2026

Cauchy integral formula, residues & series, branch points, Riemann surfaces, products, Gamma function.

Complex Analysis II.

Summer 2026

Hadamard product, Riemann mapping thm., Picard's little thm., Wiener-Ikehara thm., prime number thm., Theta functions.

Ergodic Theory.

Winter 2024

Simon Rubinstein-Salzedo. *Ergodic Theory*, The Carus Mathematical Monographs. MAA Press 2025.

Differential Geometry.

Spring 2022

Andrew Pressley. *Elementary Differential Geometry*. Springer, 2001.

Markov Chains.

Fall 2020

D.A. Levin et al, *Markov chains and mixing times*. American Mathematical Society 2006

Representation Theory

I haven't taken a formal course, but have studied the subject through the following books.

Introduction to Lie algebras.

2006

Karin Erdmann. Springer Verlag, London.

Introduction to Lie Algebras and Representation Theory.

1972

Humphreys, J.E. Springer Verlag, New York.

Representations of Semisimple Lie Algebras in the BGG Category $\mathcal{O}$.

2008

Humphreys, J.E. Graduate Studies in Mathematics **94**, AMS.

Lie Superalgebras and Enveloping Algebras.

2012

Ian. M. Musson. Graduate Studies in Mathematics **131**, AMS.

Yangians and Classical Lie algebras.

2007

Alexander Molev. Mathematical Surveys and Monographs **143**, AMS.

Dualities and Representations of Lie Superalgebras.

2012

Cheng, S. and Wang, W. Graduate Studies in Mathematics **144**, AMS.

Lectures on Symmetric Tensor Categories.

2021

Pavel Etingof and Arun Kannan. [[arXiv](#)]